

# Sandeep Kumar Routray

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## EDUCATION

### Carnegie Mellon University

*Master of Science in Machine Learning*

**Relevant Coursework:** Deep Reinforcement Learning, Probabilistic Graphical Models, Multimodal Learning

Dec. 2025

GPA: 4.00 / 4

### Indian Institute of Technology Kanpur

*Bachelor of Technology in Electrical Engineering | Department Rank: 2*

**Relevant Coursework:** Data Structures, Algorithms, NLP, Digital Signal Processing, Optimization Algorithms

**Honors:** Dean's List (all semesters), Summa Cum Laude, Gold Medal for Undergrad Project

May 2021

GPA: 4.00 / 4

## PUBLICATIONS

[1] S. Routray, H. Pan, U. Jain and D. Pathak, “**ViPRA: Video Prediction for Robot Action**”, in International Conference on Learning Representations (ICLR), April 2026

[2] S. R. Dash\*, S. Routray\*, P. Varshney\* and A. Modi, “**CS-NET at SemEval-2020 Task 4: Siamese BERT for ComVE**”, in Proceedings of the Fourteenth Workshop on Semantic Evaluation, International Committee for Computational Linguistics (COLING), Dec 2020

## EXPERIENCE

### Skild AI

*Research Intern, Humanoid Team*

May 2025 - Dec. 2025

Pittsburgh, PA

- Developing a robotics foundation model using robot demos and unlabeled human videos, with efficient inference for real-world deployment

### Carnegie Mellon University | [\[ViPRA\]](#)

*Graduate Research Assistant, Prof. Deepak Pathak's Lab*

Oct. 2024 - May 2025

Pittsburgh, PA

- Adapted **multimodal LLMs** and **video diffusion models** for predicting robot actions from internet-scale human and robot videos
- Developed **ViPRA**, enabling few-shot generalization via **latent action** learning with VQVAE and **flow matching policy** for robot control
- Outperformed baselines with **16% SIMPLER** benchmark gain and **14%** real-world task improvement with minimal demonstrations
- Published at **ICLR 2026** and **Best Paper Award** at NeurIPS 2025 Embodied World Model and Decision Making Workshop

### Samsung Research | [\[CES 2024\]](#)

*Machine Learning Engineer, SmartThings Team*

Sep. 2021 - June 2024

Seoul, South Korea

- Spearheaded 3D home layout reconstruction project. Showcased at **CES 2024** and deployed across **1 million** homes globally
- Trained **ConvNeXt with focal loss** for structure detection; optimized with TF Lite and quantization for **4x** smaller, **3x** faster inference
- Enabled single-image 3D reconstruction using **NeRFs** trained on multiple synthetic views generated from a **video diffusion model**

### Vector Institute for Artificial Intelligence | [\[Report\]](#)

*Research Fellow, Prof. Sanja Fidler's Lab*

Oct. 2020 - July 2022

Toronto, Canada

- Leveraged inter-image relationships in a **Slot Attention** framework to learn object-centric features with self-supervised learning (SSL)
- Created an image context aware score function to mine positives and negative slots for **contrastive loss** to improve feature consistency
- Ablated **vision transformers** training with SSL losses on multi-GPU clusters, obtained **2 % mIoU** improvements over existing baselines

### Samsung Research | [\[Report\]](#)

*Software Engineer Intern, 6G Research Team*

May 2020 - July 2020

Seoul, South Korea

- Built an LTE scheduler using **DDPG reinforcement learning** and custom rewards to optimize throughput, delay, and user fairness
- Obtained **80% lower delay** and better **user scalability** than prevalent Proportional Fair scheduler without compromising data rates

## PROJECTS

### Simulator-based Scaling of Inference Time Compute for Robotics

Jan. 2024 - Present

- Scaling inference-time compute for robotics by combining **Chain-of-Thought** reasoning with diffusion transformer **world model** rollouts
- Attained **30%** improvement with **model-based RL** and **reward modeling** to optimize trajectory search and policy performance at test time

### Common Sense Validation And Explanation | [\[Paper\]](#)

June 2020 - Dec. 2020

- Proposed a **Siamese architecture** and **Mixture-of-Experts** with encoder based **LLMs** for efficient inter-relational information extraction
- Coupled with cross attention, achieved **94.8%** accuracy for Validation task and **89%** for Explanation task. Results published in **COLING '20**

## SKILLS

**Languages:** C, C++, Python, MATLAB, SQL

**Technologies:** Docker, Git, PyTorch, JAX, CUDA, ONNX, TF Lite, NetworkX, OpenCV, Fast APIs